

Music Playback II

Week 07.



SOFTWARE CLUSTER

// COMT • **COMPUTATIONAL THINKING** (CIT1C18)



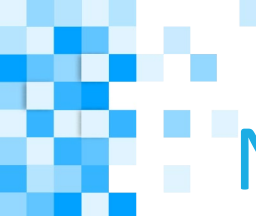
Overview

- In this workshop, you will learn how to control the player to play songs.



Objectives

- Demonstrate how to use Android **MediaPlayer**;
- Demonstrate how to write conditional statements;
- Demonstrate how to write methods and parameters.



Note

- Throughout this document, the acronym “**app**” will be used to refer to **mobile application**.

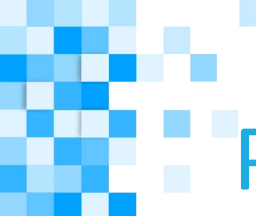


PLAYING STREAMING MUSIC

Playing Streaming Music

- In this lesson, you will be getting the music to stream and play.
- In the previous lesson, you built the playback screen as shown here, and set the `onClick` action for the **PLAY** button as `playOrPauseMusic`.
- You will be going to code the `playOrPauseMusic()` method so that when the button is clicked, the method is executed.
- But before doing so, you will need to prepare the media player.





Playing Streaming Music

- There are some specific steps to prepare the media player. These steps are documented in the Android documents.
- Here, you will learn the basic steps to prepare the media player as described on the right.

1. Create a new `MediaPlayer`.
2. Set the Audio Stream Type to be `STREAM_MUSIC`.
3. Set the source of the music.
4. Prepare the player for playback.

Playing Streaming Music

- The codes for `playSong()` method in `PlaySongActivity` is shown below.
- Add this code in `PlaySongActivity.java` and call this method in `onCreate()` method (next slide).

```
public void playSong(String songUrl) {  
    try {  
        player.reset();  
        player.setDataSource(songUrl);  
        player.prepare();  
        player.start();  
        //gracefullyStopsWhenMusicEnds();  
  
        btnPlayPause.setText("PAUSE");  
        setTitle(title);  
    } catch (IOException e) {  
        e.printStackTrace();  
    }  
}
```

Preparing the mediaPlayer and inputting a valid url so that it can start playing

The "try" and "catch" clause is mandatory for manipulating the mediaPlayer

Another method that will be used later and therefore commented out.

Setting the text on the btnPlayPause button to be "PAUSE" since we know that the music had been started

Setting the text on the actionBar to reflect the song title. Here we are accessing the class variables which we had previously created

Playing Streaming Music

- Next, uncomment the highlighted line of code in `onCreate()` method as shown below.
- The song should start playing but you will have no way to pause or stop it. The only ways are to allow the song to finish playing or to stop your app in the emulator. This will be addressed later.

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_play_song);
    btnPlayPause = findViewById(R.id.btnPlayPause);
    Bundle songData = this.getIntent().getExtras();
    currentIndex = songData.getInt( key: "index");
    Log.d( tag: "temasek", msg: "Retrieved position is: " + currentIndex );
    displaySongBasedOnIndex(currentIndex);
    playSong(fileLink);
}
```

Playing Streaming Music

- Now, you will code the playback button action to pause/resume the music
- The name of the method for the playback action is `playOrPauseMusic`.

```
public void playOrPauseMusic(View view) {  
    if (player.isPlaying()) { //if player is playing  
        player.pause();  
        btnPlayPause.setText("PLAY");  
    } else {  
        player.start();  
        btnPlayPause.setText("PAUSE");  
    }  
}
```

A simple if else statement to check if the mediaPlayer is currently playing.

If true, then pause the mediaPlayer and set the text on the playOrPauseMusic button to "PLAY".

If false, then resume the mediaPlayer and set the text on the playOrPauseMusic button to "PAUSE".

The playOrPauseMusic button should already be wired up to call this method when it is being tapped

Try out your app. You should be able to play the selected song and also able to pause/resume the music

It is very important to try out your app whenever you make small incremental changes.

Detecting When Music Ends

- What happens when the music ends? How does the player detect that the music has ended?
- We will teach you how to do so here.
- The codes below will get the player to detect when the music ends.

This is an Android api that will keep track of the mediaPlayer playing status. It will execute the onCompletion method when the mediaPlayer completed playing a music. You will decide what you want to do in the method

```
private void gracefullyStopsWhenMusicEnds() {  
    player.setOnCompletionListener(new MediaPlayer.OnCompletionListener() {  
        @Override  
        public void onCompletion(MediaPlayer mp) {  
            Toast.makeText(getBaseContext(), text: "The song had ended and the onCompleteListener is activated\n" +  
                "The button text is changed to 'PLAY'", Toast.LENGTH_LONG).show();  
            btnPlayPause.setText("PLAY");  
        }  
    });  
}
```

Here, we are doing 2 things.

- 1) Toast out a message (for checking) and
- 2) Change the text on the playOrPauseMusic button to "PLAY" since the music had stopped.

The grey color does not imply error but simply to inform you that the method or variable that was created was not being used.

Detecting When Music Ends

- After you place the codes inside the `gracefullyStopWhenMusicEnds` method, you would need to call this method whenever you start to play a music so that the `EventListener` will become active

```
public void playSong(String songUrl) {  
    try {  
        player.reset();  
        player.setDataSource(songUrl);  
        player.prepare();  
        player.start();  
        gracefullyStopWhenMusicEnds();  
  
        btnPlayPause.setText("PAUSE");  
        setTitle(title);  
    } catch (IOException e) {  
        e.printStackTrace();  
    }  
}
```



Summary

- You have now created the play/pause/resume function!
- In the next lesson, you will learn how to play the next or previous song when the respective buttons are clicked.



Supplementary Materials

Topic	Source
Java Method/Function	https://www.learnjavaonline.org/en/Functions https://mathbits.com/MathBits/Java/Methods/Lesson1.htm http://www.horsesnw.com/articles/JavaMethod.htm
Java Conditionals	https://www.learnjavaonline.org/en/Conditionals
Android MediaPlayer	https://developer.android.com/guide/topics/media/mediaplayer.html http://www.androidhive.info/2012/03/android-building-audio-player-tutorial/ https://www.tutorialspoint.com/android/android_mediaplayer.htm

THE END

